

Overview and Comparison of Several Relational Database Modelling Methodologies and Notations

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Abstract – The purpose of this work is to introduce and compare several relational database modelling methodologies and notations and to show their application through an example. The concepts, methodologies and notations are defined and described in order to compare them mutually. Moreover, a database model is shown for all five notations: IDEF1X, Crow's foot, Chen, UML, and MIRIS.

Keywords: data modelling, database, methodology, notation, IDEF1X, Crow's foot, Chen, UML, MIRIS, entity, relationship, cardinality

I. INTRODUCTION

Information technology is developing very fast, with information systems being more and more complex and the demands getting bigger. The importance of relational databases has increased with the development of hardware, the appearance of the Internet, electronic stores, business intelligence, mobile communications, etc. Today, relational databases are at the root of information systems and one of the main components of software engineering.

The first step in creating a relational database is collecting the necessary data and its analysis, which is followed by semantic modelling or, in other words, building a conceptual model. The third step is logical modelling – that is, transforming the conceptual model into a relational model. The final step is the physical implementation of the relational database.

The semantic or conceptual modelling is a process that visually represents a data model using certain methods. The method “entity-relationships” (ER) is a graphical representation of a conceptual structure of a problem domain being modeled [21]. ER is a semantically rich data-modelling method because it deals with human-friendly concepts [2]. Concepts that are used in ER diagrams are: entity, weak entity, association, weak association, attribute and relations. An entity is an object that exists and is distinguishable from other objects. Weak entity is an entity that cannot be uniquely identified by its attributes alone, therefore, it must use a foreign key in conjunction with its attributes to create a primary key. The

foreign key is typically a primary key of an entity it is related to [22]. An association defines a relationship between two entities based on common attributes. The association allows entity objects to access the data of other entity objects through a persistent reference [23]. An attribute is a characteristic or trait of an entity type that describes the entity [24].

Various notations and methodologies for relational database modelling are used daily, but this work will deal with the IDEF1X, Crow's foot, Chen, UML and MIRIS notation.

As for the paper organization, section 2 gives an overview and a relational database model example of each notation that is mentioned in the introduction. Section 3 shows a graphic comparison of the notations and describes the main differences between them. Also, that section describes a subjective comparison of notations based on criteria that is defined one chapter before. Section 4 holds a conclusion.

II. RESEARCH METHODOLOGY

In the continuation of this work, there will be a description of the methodologies and notations in order to make the later comparisons as clear as possible. A relational database model has been built for each notation using the Microsoft Visio tool [3] so that the differences between them would be additionally highlighted.

Microsoft Visio supports all those notations, except MIRIS but because it contains template for Chen notation and other general symbols it is, also, simple to draw database model in MIRIS notation. Besides, according to Capterra.com Microsoft Visio is ranked high with an overall grade of 4.5 from 5 [25] and it is free for students within the Academy program.

Figure 1 shows a graphic comparison of the notations based on the elements: entity type, weak entity, association, weak association, attribute, generalization, primary key, foreign key, and various cardinalities. The figure distinctly shows that the notations IDEF1X, Crow's foot and UML are similar and so are Chen and MIRIS.

	IDEFIX	Crow's foot	Chen	UML	MIRIS
Entity type					
Weak entity		No element		No element	
Association	No element	No element			
Weak association	No element	No element	No element	No element	
Attribute					
Generalisation					
Primary key	Attribute (PK)	PK Attribute	Attribute	<<PK Attribute >>	Attribute
Foreign key	Attribute (FK)	FK Attribute	Attribute	<<FK Attribute >>	Attribute
Cardinality: zero to many					(0,M)
Cardinality: one to many					(1,M)
Cardinality: one to one					(1,1)
Cardinality: zero to one					(0,1)
Cardinality: one to number		No element			No element

Figure 1. The comparison of notations according to chosen elements

The most of concepts from Figure 1 are used in model. Entities Chef, Ingredient and Recipe are strong entities, Item is weak entity which depends on Recipe entity, and Cooking is associative entity. All entities have attributes and relations that are used between entities have different cardinalities. Because of simplicity of the model, only the hierarchical relations (generalization and categorization) are not shown.

After building all the relational database models, concepts of each notation were compared, initially descriptive, and later in the table. The last section is intended for subjective comparison and grading based on defined criteria.

A. IDEF1X (Integration DEFINition for information modeling)

IDEF1X [4] is a relational database designing method. It is based on the ER approach for semantic data-modelling developed by Chen [5] in 1976. It contains many concepts which can be found in ER models such as: entity, attributes and relationships between entities.

Entities are represented using rectangles with curved or non-curved corners. Entities with non-curved corners can exist without any other entity – that is, they do not depend on any other entity in the model. Weak entities have

curved corners and they are dependent on another entity in the model. The name of the entity appears outside the rectangle while the attributes are listed on the inside. The attribute name must be unique in the whole model and the meaning of the name must be consistent. Each attribute can belong to only one entity (one-owner rule) and each attribute has to have a value (no-null rule). Moreover, no attribute may have more than one value (no-repeat rule). Some attributes are also keys. Each entity must have at least one key and that key is written in the first part of the rectangle separated by a horizontal line from the rest of that entity's attributes.

Like entities, relationships can be dependent or independent. The dependent relationships are marked with a full line and they are used when that entity depends on another entity and its primary key is comprised of that other entity's primary key. Independent relationships are marked with a dotted line and used for connecting entities whose primary key does not depend on another entity. If there are no markings on the relationship, that relationship is read as exactly one. If the symbol present on the relationship is in the shape of a rhombus, that relationship is read as zero or one, and if the symbol is in the shape of a full circle, then that relationship is read as zero or more. There can be the letter Z above the full circle (zero or one), the letter P (one or more) or the letter n (exactly n).

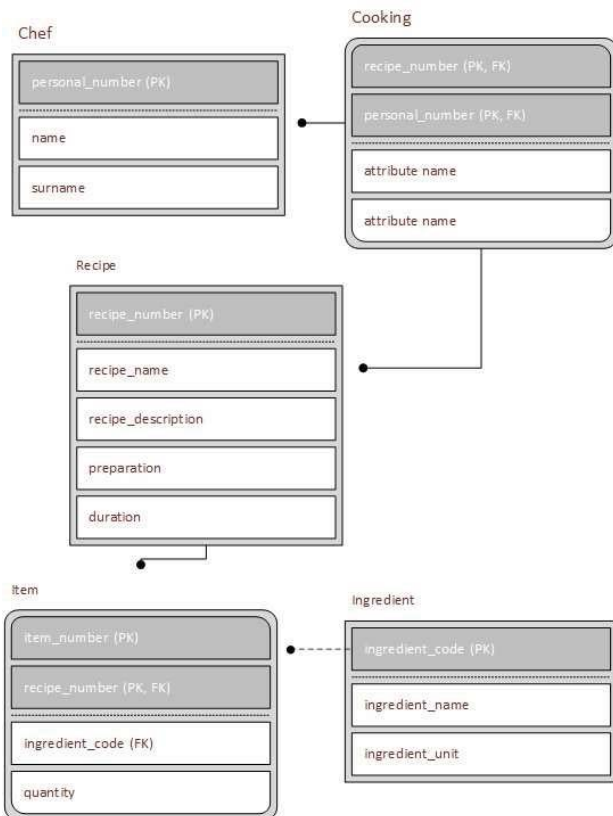


Figure 1. Relational database model in IDEF1X notation

B. Crow's foot (Information Engineering)

Crow's foot notation is one of the more popular notations for relational database modelling, maybe even the most popular. The most recognizable characteristic of this notation are the graphic symbols that mark the “more” (one or more) side of the relationship. These graphic symbols look similar to a crow’s foot, which got the notation its name.

In this notation, entities are represented with a rectangle and the entity’s name is in the upper part of the rectangle. It is recommendable that the name of the entity is singular (buyer) instead of plural (buyers). The attributes are listed in the rectangle, below the entity’s name. The first of the attributes is the primary key and it is, in most of the tools, underlined. Entities are connected using relationships. Each relationship is assigned a name that describes the function of the relationship. The relationships can be one-sided and many-sided. The one-sided relationships show the minimum number of times that an entity can be connected to another entity (zero or one). Those relationships are marked with a normal line that has a short vertical line near the entity. Many-sided relationships show the maximum number of times that an entity can be connected to another entity (one or more). Those relationships are marked with the signs > (more) and < (less) near the entity as shown at figure 2. The relationship can be optional, and it is then marked with a circle.

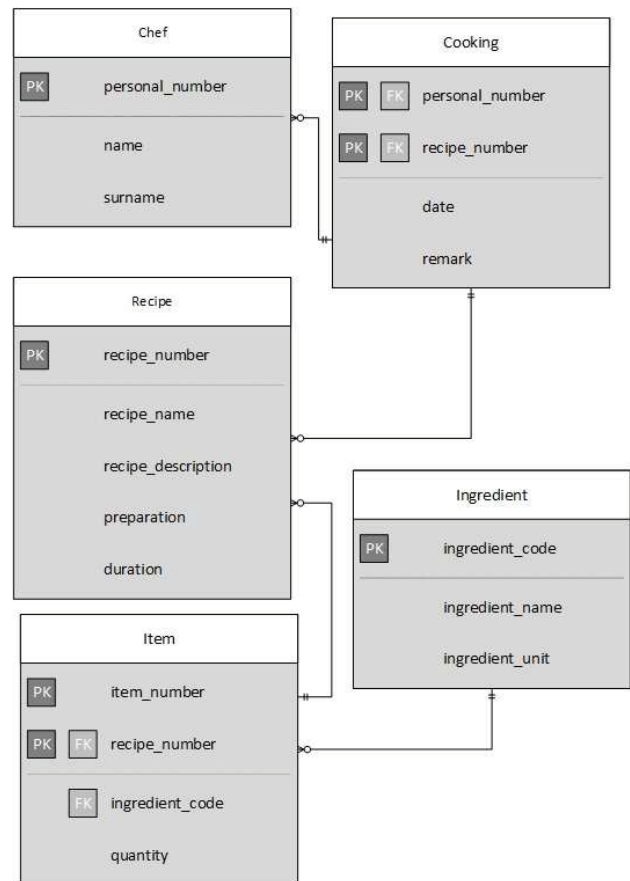


Figure 2. Relational database model in Crow's foot notation

C. Chen

The Chen notation was named after Peter Chen. In 1976, Peter Chen published a work in which he describes the “entity-relationships” model [9]. He was the creator of the concept of “entity-relationships” in information system modelling. This notation is used even today, primarily in scientific works, and is considered a more detailed way of showing the “entity-relationships” model.

In Chen’s notation, the entity is represented with a rectangle containing the name of that entity. There are three different types of entities: the strong entity, the weak entity, and the associative entity. The symbol for the weak entity is a dual-lined rectangle. The associative entity describes the relationship between two entities as one of them being a component of the other. The symbol for the associative entity is a rectangle containing a rhombus.

The attributes in this notation are marked with an ellipse which contains the name of that attribute. All the attributes of a certain entity are connected to the entity with a straight line. The attribute can be a primary key, a partial primary key, a multivalued, or a derived attribute. The partial primary key is used with weak entities and its name is underlined with a dotted line. The multivalued attribute is an attribute that can have more than one value (e.g. hobby) and it is marked with a dual line around the attribute. The derived attribute is derived from other attributes and it

doesn't have to be physically saved in the database. Such attribute is marked with an ellipse with a dotted line.

The relationships of this notation are shown with a line that has a rhombus with the relationship's name in the middle and its cardinality on the ends. The symbols for cardinality are: 1, N and M, with N and M standing for "many".

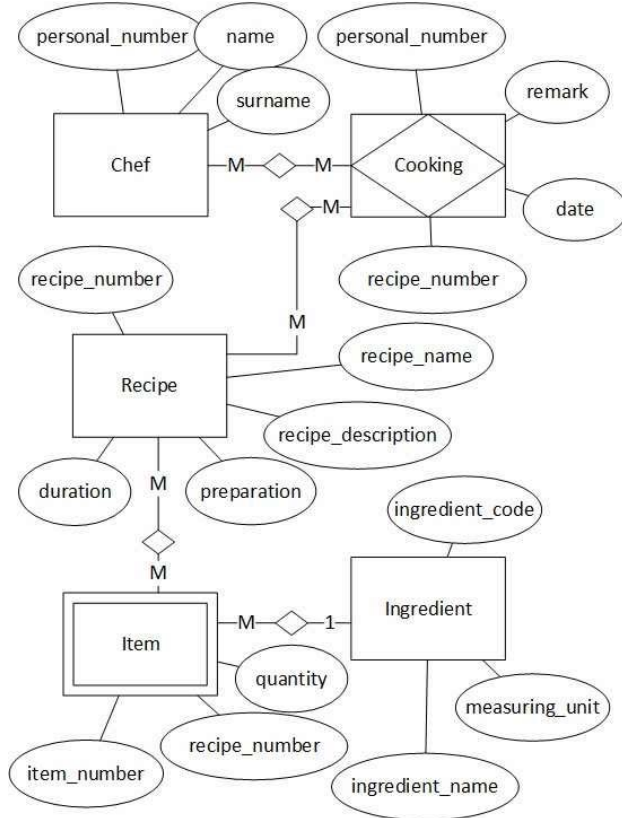


Figure 3. Relational database model in Chen notation

D. UML (Unified Modeling Language)

UML is a standard language used to determine, visualize, and document artefacts of software systems, as well as for business modelling and other non-software systems. It is accepted as a standard by the Object Management Group (OMG), and the software community has accepted it as a primary notation for object and component modelling. [7] In UML, entities are represented using a rectangle with their name and attributes written inside the rectangle. The primary key is marked by letters <<PK>>, and the foreign key by letters <<FK>> written before the name of the attribute. The relationships are full lines with the description of the relationship and the cardinality at the ends of the line. The cardinality is marked by 0..1 (zero or one), 1 (exactly one), 0..* (zero or many), 1..* (one or many), n..m (the exact range of n to m).

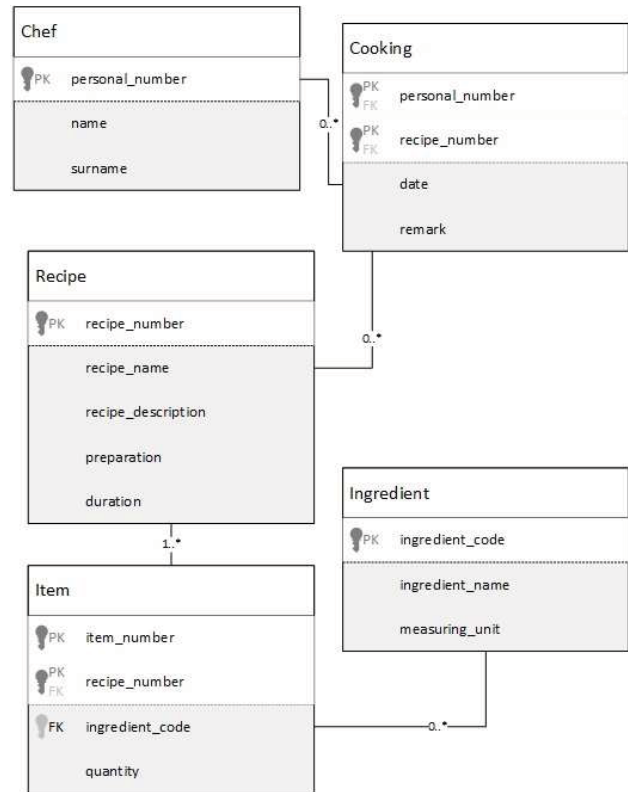


Figure 4. Relational database model in UML notation

E. MIRIS (Metodologija za Razvoj Informacijskog Sustava – Methodology for the Development of the Information System)

The MIRIS methodology is a collection of methods and instructions with the collective goal of designing and building an information system. The life-cycle phases are divided into two groups: the logical shaping (information system designing) and the physical shaping (information system building) [2]. The logical shaping consists of the phases of strategic planning, the main design and the detailed design, while the physical shaping consists of the programming support implementation phases: its introduction, application and maintenance.

When it comes to notations, the MIRIS methodology is almost the same as the Chen notation. Entities are rectangular - shaped and attributes are represented by an ellipse. The relationships are indicated with a full line, showing the cardinality on either side. The types of cardinality are: (0,M), (1,M), (1,1), (0,1).

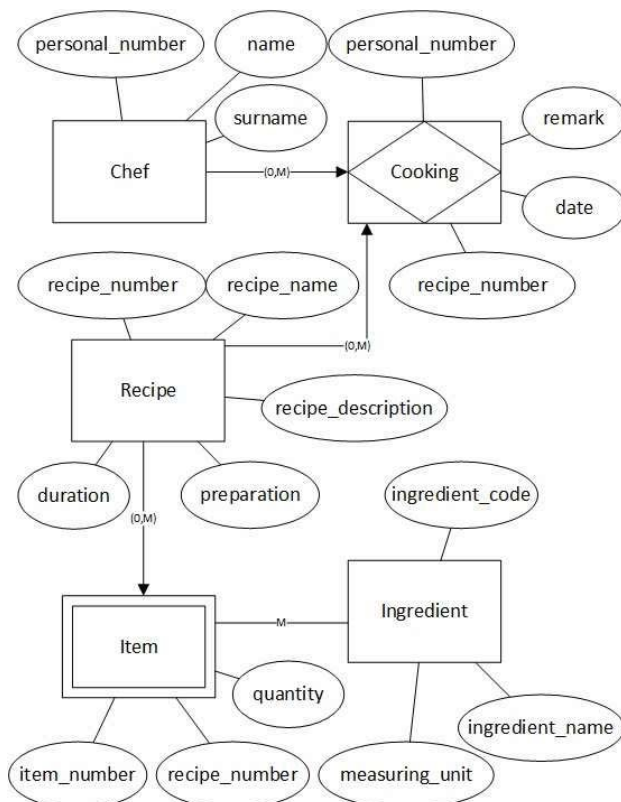


Figure 5. Relational database model in MIRIS notation

III. RESEARCH RESULTS

In all the notations, entities are represented by rectangles. The IDEF1X notation is the only one that has the entity name outside the rectangle, while all the other entities have the entity name inside the rectangle. There is no distinctive way to represent a weak entity in Crow's foot and UML and in IDEF1X, the weak entity is the same as the strong entity except that it has curved corners. Chen and MIRIS represent the weak entity with a double-lined rectangle and they are the only notations with a distinctive representation for association, which is represented the same way in both notations: a rhombus inside the rectangle. The attributes of these notations are written inside ellipses around the entity itself, which makes it less straightforward than it is in the other notations which all have the attributes listed inside the entity rectangle. Association in UML is represented by a rhombus on the relationship. Weak association is an element that only MIRIS distinguishes and that is the main difference between the MIRIS and Chen notation. When it comes to primary and foreign keys, there are no major differences between the notations. Attributes that are keys in Chen and MIRIS notations are underlined (full line for primary keys and dotted line for foreign keys), while the other notations simply have the letters PK (primary key) or FK (foreign key) to mark the keys.

The biggest differences between the notations are the cardinality of relationships. IDEF1X represents the

cardinality with a circle next to the corresponding letter, Crow's foot uses symbols which are similar to a crow's foot and the other three notations mark the relationships in a classic way, with a line that has a number, a letter or an asterisk as a way of showing cardinality. In the IDEF1X notation, the cardinality is not obvious. The letters which stand as a representation of cardinality are not noticeable and in some tools for building relational database models, they are not even supported. When it comes to cardinality, MIRIS has exceeded the other notations with its intuitiveness and simplicity of representation.

Table 1 shows the comparison and the assessment of the notations based on speed of drawing, intuitiveness, simplicity and support in relational database modelling tools. The maximum score for each criteria for one notation is 5 because every criteria, except support, is tested by 5 individuals. Five persons have drawn models in the given time. For each person who finished with drawing model in given time there is one point for speed of drawing. The same persons evaluated intuitiveness and simplicity. After drawing they chose "yes" or "no" for intuitiveness and simplicity for every notation. The number of "yes" is grade for those criteria. Regarding to support, support is tested in 3 different modeling tools: MS Visio [3], Draw.io [26], Creately [27]. Those tools support IDEF1X, Crow's foot and Chen notation. UML is supported, in some tools through UML Class diagram, and that is why it has one point less than maximum. MIRIS is not supported in any of listed tools but it can be easily drawn with modification of Chen notation.

TABLE I. COMPARISON OF THE NOTATIONS

Tool name / criteria	IDEF1X	Crow's foot	Chen	UML	MIRIS
Speed of drawing	5	4	3	5	3
Intuitiveness	4	2	4	4	4
Simplicity	5	2	3	4	3
Support	5	5	5	4	2
Sum	19	13	15	17	12

When it comes to speed of drawing, Chen and MIRIS notations are the slowest, unlike IDEF1X notation. Drawing the ellipse for each attribute in a model greatly slows down the process of drawing and makes it less straightforward and simple. Some entities can have more than ten attributes, because of that the models drawn in those notations are too crowded and look unprofessional. The cardinality in Crow's foot is different than any other notation. It uses symbols which are similar to a crow's foot versus other notations that mark the relationship in a classic way. For that reason, drawing models in Crow's foot is complicated and better to be used by the professionals not beginners. The UML notation is quite intuitive and easy to draw, but UML is intended for a much wider field and because of that, some of the elements are left undefined.

IV. CONCLUSION

In this article, five notations of database modelling have been compared: IDEF1X, Crow's foot, Chen, UML, and MIRIS. Before comparison was made, description, detailed overview and database model for every notation has been presented. All the models were built in the same tool so the differences between them would be easy to spot, but also that the simplicity, intuitiveness and speed of drawing each model could be subjectively compared.

Notations were compared based on the concepts presented in figure 1. Although, that figure shows that the notations IDEF1X, Crow's foot and UML or Chen and MIRIS are similar, grades in table 1 for similar notations are different. It is important to note that grades in table 1 are given based on a subjective judgement.

According to the criteria and grades defined in table 1 the first choice would be the IDEF1X notation. This notation achieved the highest grade with the overall score of 19 from a maximum of 20. It is intuitive, simple to use and supported in nearly every database modelling tool. The second place is given to UML notation. Because UML is intended for a much wider field, some of the elements are left undefined. In this case that is a minus, but on other hand more people are familiar with that notation and database modelling tools. With one point less, on the third place, is the Chen notation. This notation is not that simple and fast to draw and that is the only negative thing about it. The last place is shared by Crow's foot and MIRIS notation. According to subjective opinion, because of the very low overall score, those notations are not recommended.

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